

Supplementary Material

Finite-time perturbation selection in evolving HCP crystal
plasticity:
path-dependent direction, wavelength and mechanism

1. Complete fixed-candidate norm-selector screen

Figure 1 reports the complete terminal 11-by-7 screen that was previously summarized by representative columns. Each cell gives the winning log gain and abbreviated branch label among the three registered onset- x , onset- y and terminal- y candidates. This figure is a fixed-candidate sensitivity screen; it is not the re-optimized result in Section 2.

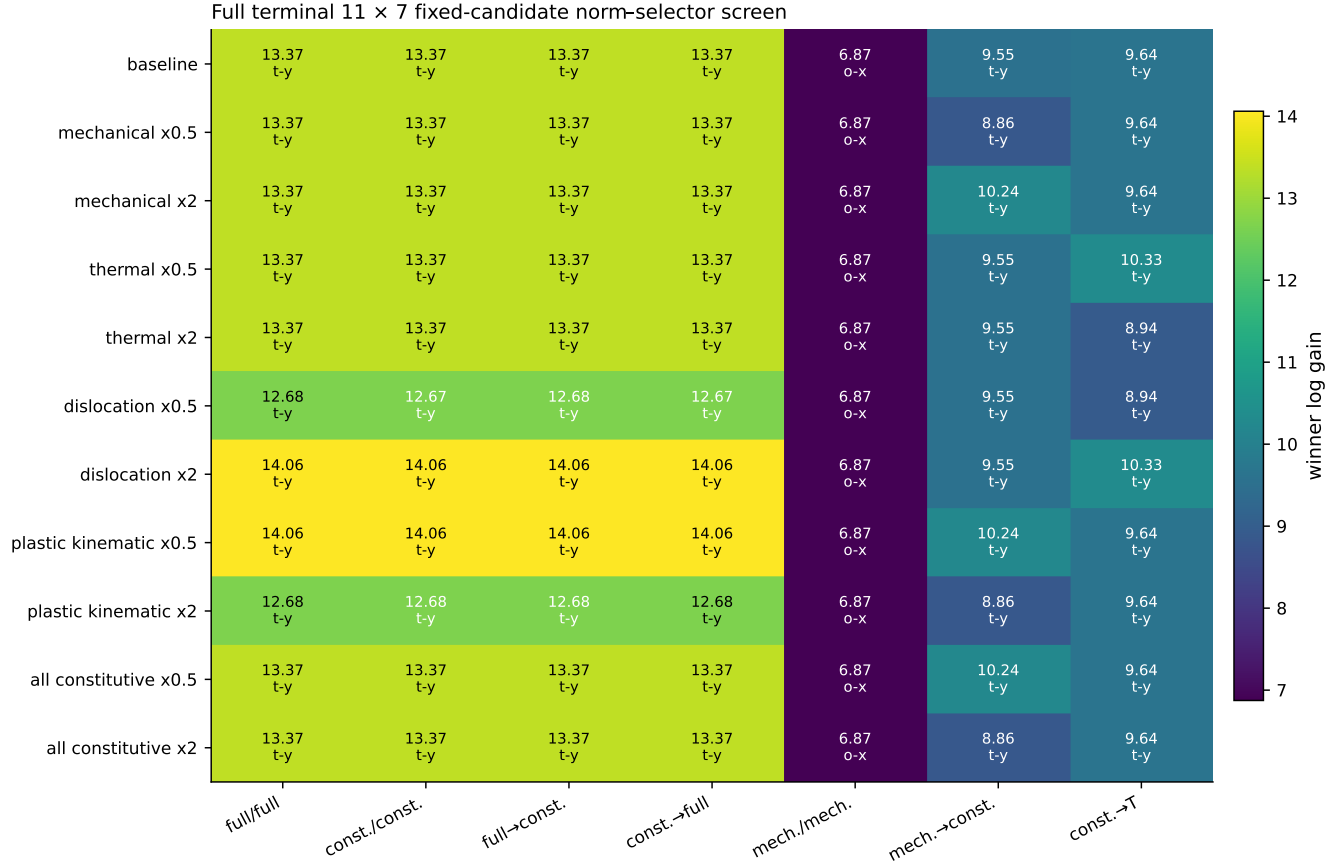


Figure 1: Complete terminal fixed-candidate norm-selector matrix. The rows are the baseline norm and ten factor-two coordinate-scale perturbations; the columns are all seven declared input-output maps. Cell annotations report log gain and the winning registered branch.

2. Re-optimized direction–wavenumber and matched freezing

Direction and wavenumber are re-optimized for all seven selectors under the baseline and four pre-registered adverse norms. The singular-value audit applies the formal 5% relative-gap threshold, and the piecewise-frozen audit uses exactly the same metric and endpoint selectors as the time-ordered map. Machine-readable coordinates, gains, complete singular values/vectors, baseline generators, convergence rows and all optimizer evaluations are supplied in `05_results/ijp_revision_evidence_v3/` and `05_results/ijp_revision_evidence_v3_arrays.npz`. The synthesized figure is included in the main paper rather than duplicated here.

3. Loading transfer, local sensitivity and titanium landmarks

The V3 table directory supplies the three added loading cases, the branch-local $\pm 20\%$ sensitivity records and logarithmic derivatives, and the bounded external check against synchronized Grade-II CP-Ti stress/temperature landmarks. The latter is a timing-and-scale check, not a batch-calibrated validation. The figure synthesized from those tables is included in the main paper.

4. Reproducibility contracts

The complete V3 receipt is `05_results/ijp_revision_evidence_v3.json`. The exact literature landmarks and extraction record are under `04_reproducibility/literature_digitization/guo2019_prl/`. Parameter source keys S4 and S5 identify the exact paper, frozen kernel version, public repository commit, configuration paths, SHA-256 checksums and access date.